

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (2 hours) March 30th, 2019

Instructions: Attempt any three out of the following four questions. Mention which three you would like to have graded. Open book and notes, plus non-programmable calculators can be used. No computers, programmable calculators, cell phones or other communications / web accessing devices.

Put Your Name Both on the Blue Book and the Test Sheet

Problem 1. Let $X_1, \dots, X_n$ be i.i.d. with pdf

$$f(x|\theta) = \theta x^{-2}, \quad 0 < \theta \leq x \leq \infty.$$

- (1) Find a sufficient statistic for θ .
- (2) Find the MLE and the method of moments estimator of θ .

Problem 2. Let $X_1, \dots, X_n$ be i.i.d. with pdf

$$f(x|\theta) = \theta x^{\theta-1}, \quad 0 \leq x \leq 1, 0 < \theta < \infty.$$

- (1) Find the maximum likelihood estimator $\hat{\theta}_n$ of θ .
- (2) Evaluate $E\hat{\theta}_n$ and $\text{Var}(\hat{\theta}_n)$. Is $\hat{\theta}_n$ unbiased? Is $\hat{\theta}_n$ consistent?

Problem 3. Let Y be a random variable that has a Poisson distribution with parameter λ and suppose that λ is also a random variable with a Gamma distribution with parameters $\alpha=2$ and $\beta=1$.

- (1) Find the joint distribution of Y and λ .
- (2) Find the unconditional marginal distribution of Y and calculate $P(Y=1)$.

Problem 4. Suppose that A is a $p \times p$ nonsingular covariance matrix with eigenvalues $\{\lambda_1, \dots, \lambda_p\}$ and eigenvectors $\{v_1, \dots, v_p\}$ and let $\{x_n\}$ be a sequence of p -dimensional vectors defined as follows: $x_{n+1} = A x_n / \|A x_n\| = A^n x_1 / \|A^n x_1\|$ for $n = 1, \dots, \infty$. As n goes to ∞ the sequence $\{x_n\}$ converges to v_1 , the eigenvector of the largest eigenvalue of A under the condition that x_1 is not orthogonal to v_1 .

- (a) Give a similar sequence for calculating the largest eigenvalue of A and prove that your sequence converges to the largest eigenvalue of A .
- (b) How would you use the previous results to compute the second largest eigenvector and eigenvalue of A ?